



Build This IoT-Based Smart Home Controller Using Blynk

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Presented here is an Internet of Things (IoT)-based controller for a smart home. This project can be used to control any four home electrical appliances such as lights,

fans, air-conditioners and water pumps from the smartphone.

In this project, we used two light bulbs, one DHT-11 sensor to monitor room temperature and humidity, one

soil moisture sensor along with garden motor pump, and one ultrasonic sensor (HC-SR04) for water-level indication in overhead water tank. Their status, including light on/off, room temperature, soil moisture level and water level, could be monitored on the smartphone graphically.

Circuit and working

The circuit diagram of the IoT-based smart home device controller is shown in Fig. 1. It is built around NodeMCU (ESP1), two light bulbs (Load3 and Load2), four 12V single-changeover relays (RL1 through RL4), 5V regulator 7805 (IC1), temperature and humidity

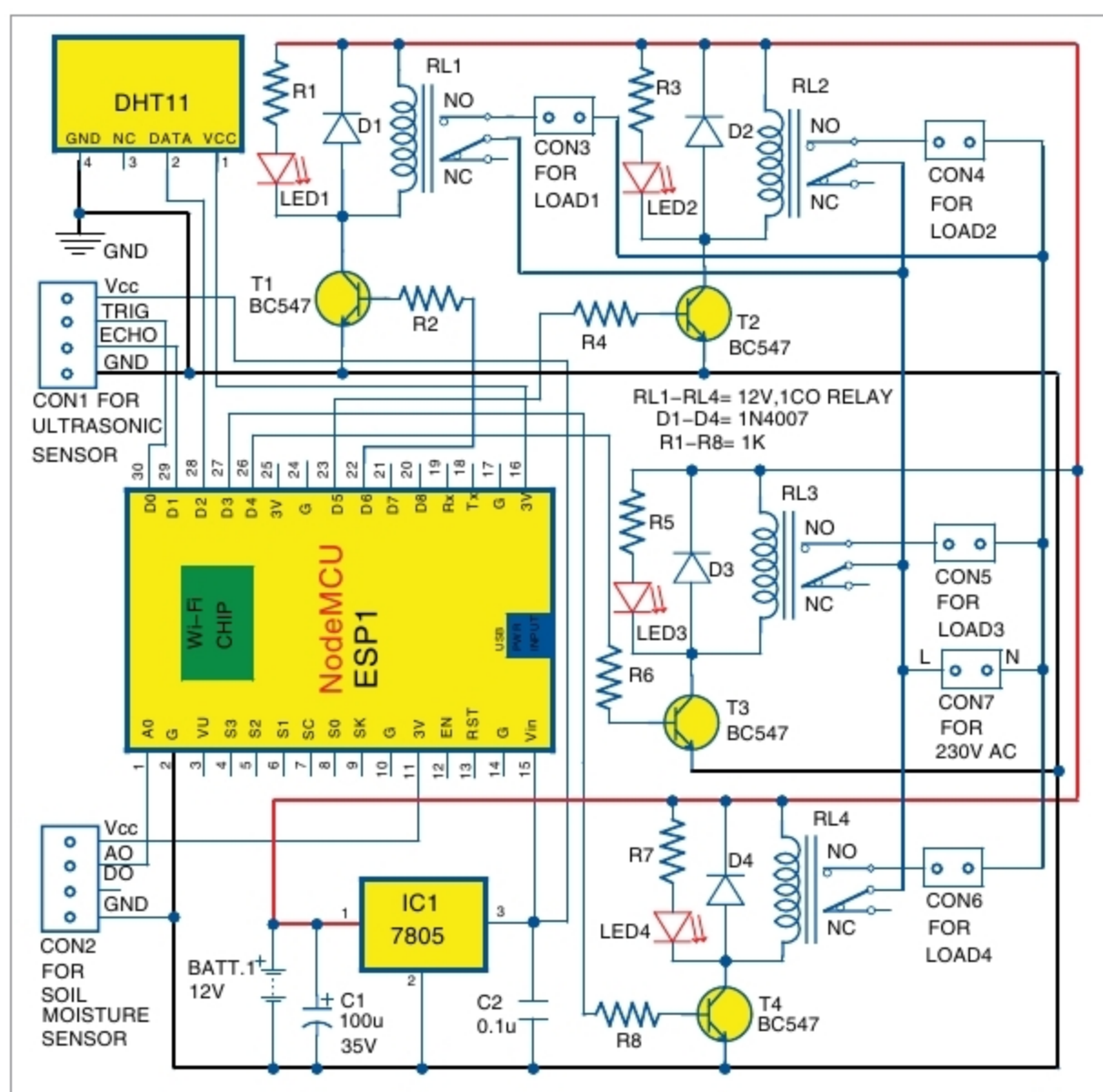


Fig. 1: Circuit diagram of the IoT-based smart home controller

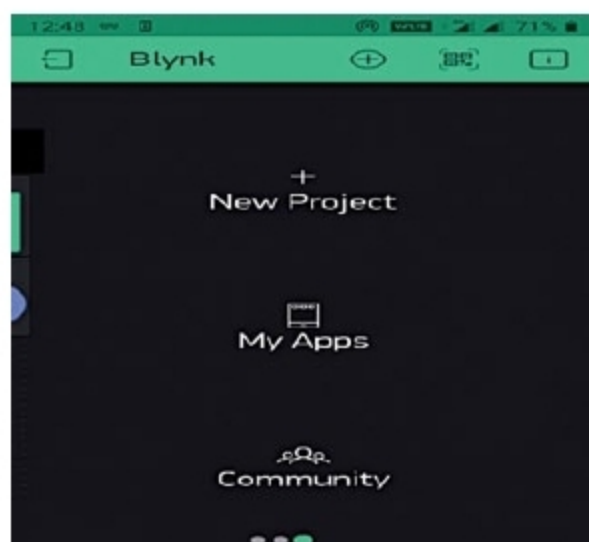


Fig. 2: Creating new project on Blynk platform

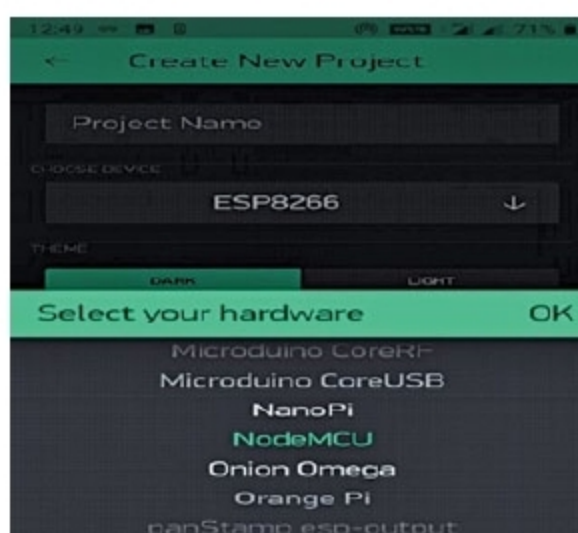


Fig. 3: Selecting module as NodeMCU

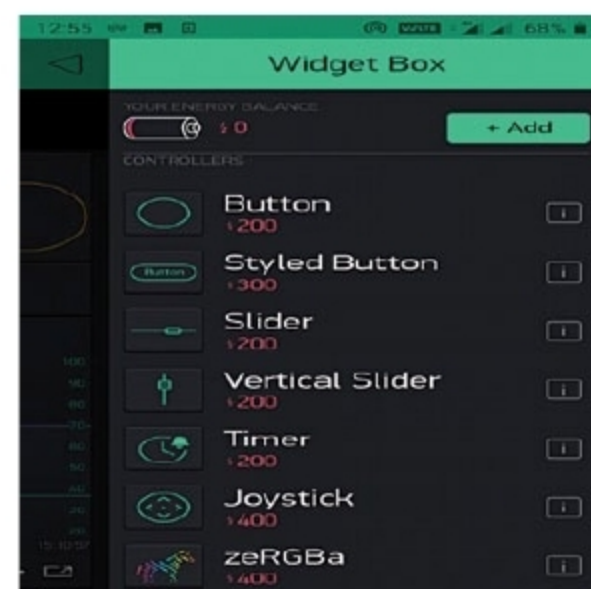


Fig. 4: Widget box to add control buttons

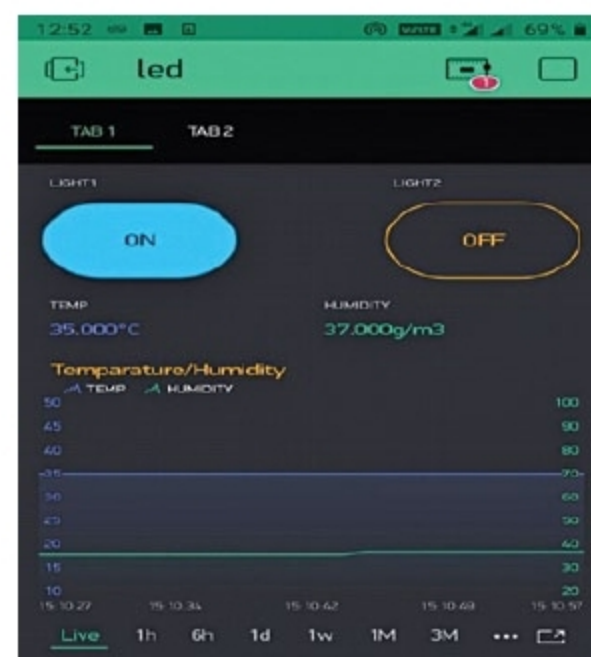


Fig. 5: Adding control tabs